

ActInf Textbook Group ~ Cohort 6 ~ Session 5 (Chapter 3, Part 1) ~ 2/26/20

TextbookGroup/ParrPezzuloFriston2022/Cohort_6 | Cohort 6 ~ Session 5 (Chapter 3, Part 1) ~ 2/26/2024 | 56:27

all right welcome

back it's March 4th the only day that's

a command March 4th so as we head into

chapter 3 does anyone want to begin with

any any thoughts or questions or remarks

about

three or where is it in the context of

the

book what do they remember from it or

what's a quote that was interesting or

just a point to jump in from

I personally found it really interesting

that the marov blanket is just an

intermediary it's just a barrier between

two things the external and the internal

State I really it reminded me of my

physics classes talking about a ball

rolling down the hill in the sense that

you could take it from the perspective

of the ball and the universe of moving

around it or the the universe is still

and the ball is moving down the hill

that freedom for that kind of reference

surprised

me yeah interesting stable Hill moving

ball and stable Ball moving

Hill that that is that it

Zach absolutely yeah yeah

that a kind of similar topic came up

yesterday in the um discussion with Ilan

Kent and Anna Unica on mental gravity on

like kind of there's a reference frame

where the Earth like spaceship earth is

the egocentric navigation that's like

our visual reality like objects coming

closer and then there's kind of a third person Sun centricism there's other reference points that can be chosen but that's definitely a very so how would you connect that to Markov blanket or to

AC or how does the Markov blanket allow us to do it this

way wait I'm sorry perhaps I don't understand are you saying they posed that question or are you asking that question oh just asking like what what made you what made you think about this ball moving down the hill from the Markov blanket

oh because of the K Divergence it's not exactly perfect but it seems as if you uh can

move X the actual like external world is going to stay still but the Q is going to be the thing that's

changing however it seems to where your choice as the ball as the thing moving down the hill is kind of arbitrary if you assume that the K Divergence is not going to change if you change up the order it was just a thought that I had that found to be interesting the kind of relative relationship between the two perhaps there's not much to it well well I'm I'm seeing

maybe the large points as being somewhat

porary what are those points so when you have an orbit um when you have two bodies in orbits there are certain points where there's an equilibrium between the gravity fields

between those two bodies so the lrange point is actually a place of stasis um and in a way because it's an absence of both those bodies and the in the influence of both those bodies it's actually a separate virtual body in and of itself so or a virtual Point excuse me that that's very interesting lock Li in in this kind of mental gravity um discussion which maybe people can look at for more detail it's like he did discuss the center of gravity those topics came up with the basian mechanics and Dalton otiva devel like you have a distribution and then you have a center of Mass center of gravity that's like the central tendency of a distribution so basing mechanics is kind of something like a gravitational interpretation of the basian updating process like big distribution heavy prior Light low attention uh sensory input not going to move the prior that much and then the um lrange point is kind of like the the stationarity if you're there like a non-equilibrium stationary Point like uh the person in the frog and the uh jumping out of the hand example a textbook they could reach like a long-term stationerity where they were just estimating something that where they kind of they've reached like a steady point with what their environment was providing so it's kind of like one Gravity from the environment one Gravity from the um agent with their prior and

then also that the um p is stable like
the true distribution of uh temperature
stable Q is changing beliefs are
changing or beliefs are stable true
distributions are
changing and that kind of uh maps to
the um change the world change the
Mind
possibility
like belief
updating and fixed belief World State
changing so nice lot of great comments
here

Thomas and then

yeah yeah two points one is I found
really fascinating the Reliance on
temporal dimensionality for a lot of the
argumentation here so for example the
argument of basic the high rical gener
generative model for example the
illustration of sentences right composed
of words uh their high rical orentation
is based on a nature of duration things
are taking longer than other
uh that's a really fat basically things
are taking a rapidity a rapidness of
movement versus a duration longer period
of time um as the argumentation for like
what gives an ordering to things I think
that's incredibly fascinating another
dimension related to this idea of like
emotional like why we're talking about
this kind of psychological aspect of
gravity um there's that line where that
idea of temporal depth is directly
correlated with psychological dep of a
person and that's really interesting on
a couple grounds one is I'm attaching

this that ideas of the kind of factual
and trajectories and policies
argumentation that's been presented so
far AOS has a really fascinating line in
metaphysics about rational action
activity he uses a very particular
example of a rock a rock is IR R the
activity of a rock dropping is not
rational because it has it's compelled
by necessity to have to go down the
point versus rational actions there's
AOS a power of Choice a power of D of
deral of of either to dox or not do X so
the counterfactual point the example use
in Ariz metaphysics is uh medicine right
to heal or not to heal for example um or
that actor basically the agent throwing
the rock or not throwing the rock again
that cap like that the drawing of
rational activity to be that power of
choice of temporal optionality basically
um I saw some correspondence to notes
like that I've always always find that
very interesting AOS style he talks
about that's his def that's like
derivative his explicit definition of
what rational activity is uh which has
this kind of potentiality temporal
nature to it I thought that was really
fascinating um the other note it's a
classic problem um this is a from a from
Scholastic aquinus models of
psychology the chasm between the
fantasms and the intellectual activity
in Scholastic Traditions it's considered
a mystery they don't know what's
allowing the conversion of sense data
into the F What's called the fantasms

which I guess in core in this kind of mapping logic would be the active States uh there's no the explanation is kind of obscured how you especially the problem is our sense stum is on particulars a particular dog or a particular expression of light if you're looking at in a room for example how do we do the cognitional action of universals to get the idea of this Quantum is in front of me is the dog there's a whole thing about that in scholas atmology about that what's relevant to this group is I'm a little unconvinced that this argument of the blanket is solving that problem I say that because there what exactly is the C like so if if there's like three variables that are basically being interchanged with active States external States sense States and internal States you know obviously the correspondence of the variables between active States and external States and sense states to internal States totally get it like I could see how this Maps right uh but between the variable X and Mew what is that c like how is that connected I'm not I didn't see anything in the documents in the literature so in this chapter so far to give that explanation of how that conversion is happening uh I might I may have Mis read that I might completely be a mistake but this is just in my mind I just know this is a traditional uh problem in epistemology of basically the application of universals of perception

and cognitional activity there's a no
one really knows how to explain give an
account for that so that's why this
chapter caught my attention and I'm a
little confuzzled about this uh because
eventually when he get drawn down to be
uh yeah unit U Bas of the variables
become U and Y when really μ and X are
seem to be the question mark of what
that is the thing of of translation uh
between the sense internal States and
the active external States how is that
happening I don't see that happening yet
um okay those those are just my
generally notes again I might be widely
wrong about this so I'm happy to be
corrected on that more strongly stated
second point but also that first point
on specifically the temporal nature of
of arrangement of generative hierarchies
and synchronicity coming in is a very
important factor I saw I read that I
actually have that down in my notes um
let's see what was
that yes that the internal states have
an appearance of representing external
States so μ dot is representing
appropriately x dot by the synchronicity
of the active uh state in the sense
statey and
do that's what I got thank you
thank you for the great comments so to
the second one first that's a very
perceptive comment which is that the um
particular
partition which is the starting point
for a free energy principle active
inference and not necessarily starting

point to get to any
phenomenological or metaphysical Point
potentially only as a starting point
just to do something empirical so first
a failure or a Delta to a philosophical
explanation may just be like a permanent
open question for a lot of other reasons
so the but but just to the kind of
philosophical implication part the edges
which represent what basically can
happen it's basically all the edges can
happen except for two kinds of things no
backwards
action no backwards perception but the
other arrows are like in principle
double-sided that doesn't mean in any
given situation they're like of equal
relevance also this is a map not a
territory so like just drawing a line
here between um you know the the the
North and the South side of the city
that that that line on the map is like
it's literally not describing what the
materiality is of the connection so
that's going to come back like again and
again obviously the map not ter
question so first category thing that
doesn't happen that reverse sensation I
don't have my thumb on the scale of
sensation or it's modeled like um part
of the sparsity of the models that it
just can't happen and the other thing
that can't happen is that action can't
be controlled externally action is under
um one the autonomous
control this is not like just pick it up
and pce it on to like an agent as
complex as a human because as we'll get

into like even in a tze we'll see how
limited it can be so this is just a
starting point of kind of like the the
like OnePlus One Lego block that then
more sophisticated models especially
with internal sophistication RS um but
the other kind of thing that doesn't
happen other than the backwards action
backwards sensation is the no telepathy
no no um telekinesis I guess like no no
communication from the outside except
through sensation no action effect on
the ex outside out except for through
action so that kind of gets this
question of like um how but isn't what
we actually care about the coupling
between the systems which is like the
The Edge that would connect these two I
think the um

[Music]

um it it's not with the Conor hind was
26 um basing Mechanics for stationary
proes that kind of comes back to this um
Irange
point this one talks about the blanket
so here's before partitioning the
blanket out
into um incoming sensory and outgoing
action so this is just the the um
initial marov blanket concept on an
undirected graph so both of these edges
are biral so this is kind of the simpler
version of
um this where the blank kit is broken
out into
two then um they study this the Locking
of the
mapping the actual stationarity in the

tracking is between the internal and the external

state that is one that is the state S_t
it that is the kind of good regulator
like

stationarity so it captures both the um
intervention of the blanket

State and the I I'm I just don't
remember exactly which paper has the the
sigma

mapping between maybe someone else
remembers but the sigma mapping is is
the one that describes um

between the internal and the external
state so there is like it's like an
image or a map and it Maps between the
internal and external States but it's a
but it's a it's another um projection
but this only describes the causal uh
connections

Jeff yeah I just I just had a thought um
that I really haven't seen much in the
literature um until very recently about
Magneto

reception and also we know that we're
also sensitive to gravity but I haven't
really seen that modeled as a sense that
we actively participate in as well so
maybe there's something there when
you're talking about so-called
telekinesis um where where it's actually
just a natural phenomenon that that that
may be because we know for example Birds
right Birds navigate using the Earth's
magnetic field we know that um we know
that different different types of
animals have different sensitivity to
gravity fields we know that when for

example when we go Cuba Diving we lose
our sense of gravity right um so maybe
maybe that's just something to to just
stick in your your file cabinet I don't
know yeah definitely this live stream
from yesterday it was a rare Sunday live
stream like it it links to neurobiology
and a lot of other things and we we we
couldn't think of a linguistic example
that didn't
convey um that wasn't concordant with
like a gravitational
like metaphor being applied in a certain
way

Zach uh hi uh so can you say a little
bit about so there's no telekinesis yeah
but what if control over one specific
variable is in like one Markov blanket
and another is in another Markov blanket
would the separate Markov blankets under
some other hierarchical Markov blanket
be considered an external state or would
those all be inside of one big blanket
that you just call the set of internal
States if that makes
sense yeah

um though in this like minimal form
they're labeled this
way they're all relative to each other
and that's all relative to the choice of
which one is chosen by the map drawer as
the internal state so it's not like
aspects of the world are like oh this
one's a sensory State now we can move on
it's like in this model where this is
the internal State and this is the
external State then they have this
directionality um there's an interesting

question I think it's in it's in chapter two or three and it kind of highlights how the uh textbook talks a lot about generative model and generative process but then um it has Alli and then uh but more recently they just described the entire statistical model as the generative model so that's the way that um many people talk about generative modeling now that although there are agents that are defined by like the causal connections of the blanket in the internal States so like you can have agent based model as as you could with net logo or any other kind of modeling framework um there's no need to talk about specific variables as being intrinsically like an active state or a sensory State because if it was like a communication setting then the active for one is the sensory of the other so as soon as you're even in a a two agent setting or anything that's um like blurring the line with the basil cognition or anything you just build the program and you use the structure the syntax of the of the system that you're building the model in like we can look at pmdp and look at the code as it's kind of rendered for some people that is helpful to see that these are really just variables that are defined in their relationships um and then when you start linking them up they're they're they if you had another you know extension it's was a

question that that we hav't and I don't know if there's a specific total answer which is like okay it's a wiring diagram like this um and this was the kind of core critique or or line of um inquiry by this like how particular it's like okay the FP has these properties but does it retain those properties like as you change the topology of the ACT ction perception Niche like if you don't have this double edged Arrow or like like if you just had Around the Clock like only one two 3 four what properties does that coupling have can that one synchronize um across blanket uh so I I I don't think it's totally resolved uh but it can be explored how different connectivities and of course different parameterizations within a connectivity relate to like different functions and especially when getting into um internal states that can have um Dynamic structure so you'd say okay the temperature in the room is going to just be like it is what it is and then the um thermometer and the air conditioner kind of also are what they are but we have an agent who can create higher order abstractions about temperature through structure alert so a lot of the change happens like already blanketed off from the sensory environment which I think kind of touches on Thomas's Pro problems questions too awesome thank you so much

pretty interesting

figure cool visualization could probably
be used more broadly or like something
like that like

some you using different colors and
types oh here's the mapping here's Sigma
mapping x y and

blanket so X and Y and then sense in
action let's

see past time

Point current time

point so here's just

describing the X maps

to action and itself internal

States our action selection and have
their own recurrent Dynamics and like
okay

analogously

two Sol little

coupling um I mean we can explore these
more in the paper I don't think this is
the same q but I think this reflects
that kind of like um kind of like lrange
Point like non-equilibrium steady state
something about how they're like in a
something like they're circulating in
some way

three is the blanket condition saying
it's this set composed of sense and
action summarized with this one node
and then the

assumptions some other kind of
conditional Independence like that they
all have this one other here they look
more

connected here's the sigma
mapping here's something else about this
observations or something pretty

cool I don't know how much more graphical but but the category theory is also taking it very graphically this is still within the basian graph like visualization approach but the category Theory work probably has a lot of visualization that that's also more formal argumentation any anyone else just any other angle or we can go to some questions also people can upvote the questions but let's just look at does anyone want to look at a specific written one or new one or should we just look at the most uploaded ones Markov blanket explanation page 43 Box

31 so here's the marov blanket in time concept the or which is sometimes called the marvian property like as has kind of been discussed before the last names don't really add any information because there's like multiple marov and there's multiple senses and all these things so sometimes just using the descriptors more informative but the marvian Assumption like it just broadly in statistics like talking about like a Marian model or hidden marov model is where the past time step influences the next time step and then that influences the future time step so the past only influences the future through this kind of retained One Step present um how can this be true it doesn't have to be a true fact ontologically about a system it could

just be a statistical approximation or
designed feature like a
program what if we don't know the
present
well that's a super broad
question by even sketching out like a
few nodes on a piece of paper or
thinking a little bit about a specific
system like what like what system not
knowing what about the
presence just sketching that out moves
along that question but the general form
is probably pretty big question they
Define markco blanket well they Define
it locally but it's
also prior defined the set of variables
that mediate all statistical
interactions between a system and
environment what's the statistical
rather than a non-statistical
interaction um here maybe in the broader
sense it's saying like it's about the
statistics of the map not the actual
territory so we're already in the domain
of statistics
another um way to read it would be like
it's kind of a probabilistic like across
like many similar or comparable
examples that it's not necessarily
seeking to give like for example a
mechanistic account or constructivist
account of how one thing happens it's
giving a
observational empirical numerical
approach to modeling like categories of
events
just like anyone would doing statistics
or AI or machine learning

Thomas kind blending questions one and two together and what's the relationship of hidden states with this because I remember the model from the you know low road Mo chapter there was a discussion about hidden States being part of the basically the calculation um so just that question of not obviously the point of not having all full information is covered by the fact that there's hidden states that are being accounted for uh as I recall that from the chap on chapter two in h in the low road process so is that part of that relationship here one of the ways to rec the connect the two I questions yeah like beliefs about hidden states do come into play in the variational free energy and then also in prospective planning and action selection like in this book anytime you see X till the um that's beliefs about hidden states are always um more uncertain than beliefs about measurements so these two forms of expected free energy um expect like also always good to just kind of like threshold that not all things necessarily do this kind of policy selection process Like a Rock doesn't have to be modeled with a policy selection process or just a trivial one but then getting into this like adaptive action setting then here's like the uncertainty about how actions relate to World States is always greater than uncertainty about

how actions map onto
observations best as they're equivalent
like in chess where there's a one to one
mapping between the um observation and
the hidden state of the piece on the
board that's like a fully observable
game not to say that the opponent's mind
is fully observable like but that's the
whole modeling um
process
uh those are maybe a few aspects of it
but what else what what what are the
kind of key aspects you think and how
maybe it's been approached
philosophically otherwise
like oh are you
ask or just like stat yeah kind of this
is a statistical model but not all
thinking on this issue has been
statistical so how just as kind of open
question but maybe if you have any
thoughts of any pre- statistical
thinkers and how they approach like
boundary I mean we normally argued
by definition so the idea of an Essence
is defined by definition which is a
speci in general with the differential
basically combination um if you use mean
Plato in Theo talks about that using NE
necess necessary con property to Define
limit and thus used it to Define color
and figure and all that um so those are
non-statistical probabilities to
define basically some observation um in
fact actually from a more Norm like from
a more ethics normative point of view uh
the idea of exemplars is the basically
exact opposite that the best

representation of reality is the
exemplar versus uh an statistical
average right um so that would be kind
of the broadest kind of answer I've
agreed that broadest difference between
a statistical model and a
non-statistical is just use of
definition and Essence uh from a CL from
a traditional classic point of
view

cool really cool yeah that's awesome
thank you this is like the N equals one
and the
example and the a
priori

non-empirical and then the statistical
like or then the instrumental would kind
of which is just described basically
using it as a statistical model like
people often talk about this this is the
whole not the whole but the a lot of the
discussion around like do we do the
organism are they a generative model do
they have one are we just modeling it as
such that's this instrumentalism realism
question I mean if known claims are
being made about the territory then you
know one can always Retreat to just pure
map making but then there are stronger
guarantees of the once you've retreated
to map

making but then that has fundamentally
given up certain territory relating to
like how individual things
happen it's just and it's just a
different kind of it's a different kind
approach and and I I I know there's some
connections with the free frequentist

and the basian
statistics and like how do frequentists
view chance how does basian view chance
there's probably a lot to go in there
too yeah just to build on that just I
just by the way it wasn't like Aristotle
did not know about chance it was
discussed it's a know it was a known
thing the I think the big difference
just to CL I mean that's drawing on this
point is that
science ofis was supposed to be about
universals like this idea of kind of use
of chance to kind of observe and find
reality is that's the part where there
probably a huge Schism difference um
just from ran Point uh epistemology or
investigation there something different
so it was like they know they knew
chance existed and they definitely
acknowledged that certain things do
apply to that but it wouldn't be part of
a science it wouldn't be the mo the way
you would use to come to find the
universals is through random chance
yeah yeah and the role of abduction and
induction deduction and kind of which
ones are given what epistemic
statuses does anyone want to go to any
written question or or any new
one
here's no answer but this
Iran again that's the last the last
names points like it's it's hard to know
is this lrange point it could be but it
might be totally unrelated it could be
different person different work or it
could be exactly the

same but like a point attractor or something like that gives more information okay let's do let's just see how many from the top trying to understand better how figure 3.2 was produced EX exactly what it

means okay so there are the variables that are going to be used this is kind of the simple blanket setting there's going to be three variables X , μ and B so this is just in the undirected graph setting before splitting out sense and which is what kind of transposed it into the cybernetics context but just one blanket State u_m and then these are kind of three projections down there's a cube of statistical outcomes in the u_m like variables taking on a number in the X , Y and Z Direction but it's in the X , B and μ

directions and then here are three projections from the cube on each of the faces down onto each faces u_m just comparing two different variables u_m

so uh there's a positive relationship between just a linear aggression between X and

B and then you can condition on B like you could kind of zero out like you could talk about how many u_m standard deviations each X was away from the conditioned expectation on B so like this one would be or this one would be lower in standard deviations than this

one just be it's like the residual from

the regression so there's just statistical methods and there's other relationships that are drawn so these just describe the like conditional dependencies like kind of this this is like a statistical Edge like but that is drawn that is saying the scatter plot has a significant explanatory power um but it's not exactly tested like with a P value as such but just saying that that's the slope is like the the causal effect if the causal Edge were one then all the points would just be on a line like knowing one would be like 100% like which is one in statistic everything's kind of scaled to one everything happens on the 0 to one interval

Eric or Thomas and anyone else got to follow up on basically my my question about what what brings the what pulls the X and μ value variable together like this grap seems actually I had that in my notes I forgot talk about that right because what happens it looks like it's using the blanket B to that's the that third graph on the bottom left is the combination right of the two V of those two investigations or two measurements and then that fourth graph is the probability mapping of x given uh look at the note yeah Pro given the expected probability of μ given B conditioned on μ I think best understand to read it yeah great that that that adds a lot definitely so like

these two describe the conditional dependencies between X and μ so internal external States on blanket and then here is that mapping so it's more variable like let's just say that there's a 0.8 relationship between the temperature in the room and the thermometer noisy thermometer 80% good thermometer and then um there's a 0.8 Fidelity between me looking at the numbers on the thermometer so just coarsely it's like then there's going to be a 0.64 relationship Sigma de facto like around horn because of correlations amongst variables so yet there's no direct causal effect between the hidden State and the the external State and the internal state so that that describes both like the the intermediation and the kind of ultimate correlation and so this is this is like structural equation modeling is commonly used like in population health or the um earlier or it's oh it's in chapter four here's like some some kind of here's one hidden factor causing one observation here's one here's lightning and then here's the light part and the sound part two observations from one underlying State here's two causal factors that combine to influence one outcome these kinds of Bas and Gra iCal methods are used like from like knowledge database systems to health modeling all these different kinds of things okay if one defines these preferred States as expected

States then one can say the living organisms must minimize the surprise of their sensory observations is it that preference is the same as expectation or formally we can use them in the same equation or something else does active inference make ontological claims EG preference is actually expectation just as heat is actually excitation of molecules more generally is a free energy principle free energy in physics just an analogy or is an ontological than okay one important thing to note is that there's two senses of expectation so in everyday usage the meaning or the connotation is like what we expect to happen like what what what we expect will be the weather tomorrow however also when we see the fancy E the Big E expectation of or the sum of something divided by n data points which same as the that's also called the expectation also known as like the central tendency or the mean so the expectation of a distribution it's like the expected returns but that's a situation that's both about an uncertain future moment and it has to do with the moment with the central tendency so um it like it's just a statistic even within statistics those are just clear based on the context or when what variable um is coming to play um inactive inference preferences so four things that are choosing actions that are using their that are exerting their

preferences let's look to
chapter seven so here's a model where
there's no preferences
involved that helped calibrate some of
these statistical questions about like
the prior the relationship between the
hidden State inference and the
observations through time through the A
matrix and like the B Matrix in how the
hidden state is inferred to change
through T but there's no action exerted
in this model but that brings out a lot
of these questions about like
observation hidden State how things
change in the world all these things
like that then that bottom structure is
still exactly the same and then here's
this action selection
part that invokes this variable C which
is called
preference it's not a reward function
it's a preference function that's a
distribution over
observations so it'll again come up in
um like chapter 7 and before what what
its structure is but this is one of the
core pieces of active inference models
that the preference component of the
selection which is going to come into
play in expected free energy in the
pragmatic value so this is going to
Define pragmatic value for that
variable that is defined in terms of
observations sensory
observations let's return return to the
question see if there was more to it but
that's kind of the key piece they're not
that preferences are not about future

points preferences come into they're not
using that future um unknown outcome
sense of

expectation but you could imagine a
planning agent that also has preferences
about future States or their preferences
right now C right now comes into
play in deciding action that matters for
future events

but
that's preferences over sensory outcomes
now having influence on action
selection not like a betting Market on
what an observation is going to be at a
future time Point per

se
so is preference the same as expectation
no but it'll be especially clear when
you just even sketch out a table or a
list of what is being described because
this is a very general this is like at
the level of definitions of
words formally we can use them in the
same equation there are situations where
it's an equation where you could
describe it in English using either of
those words but there's other situations
where you you couldn't so they're not
simply

swappable for that
reason this can't be really described
also I don't I don't know about this
heat is actually excitation molecules
kind of interesting metaphor but maybe
write more about it and then let's
explore it is free energy in physics
just an
analogy or is an ontological assertion

can analogies be ontological
assertions I don't know there's probably
a lot of things say on that question but
we'll go

on what's the difference between the
terms or does anyone have another in
like 10 10 minutes or we can just look
up like a few more of these or anyone
can ask or say anything
else

okay what's the difference between the
terms uh Thomas go for it hey sorry I
mean the the idea of the generative
model hierarchical generative models
based on time could we clarify that a
little bit more because it does sound
like um the fact of a hierarchy right
basically for example again the
classical model of a of a chain of being
that thing that uh entities have a
certain orological reality or hierarchy
right that rocks are not the same as
animals animals are not the same as
humans humans and God U there's a kind
of ontological hierarchy given on that
um of some orientation of how things are
arranged this chapter seems to talk
about the time duration as that ordering
property could you get more clarity that
or just like how strong of a statement
is that or what the implications are are
that yeah greates question okay few
pieces on this so um when you brought up
the first
time let's just say that we're going to
have planning over 100 years one kind of
model would be like a model where every
year from 1 to 100 is represented like

on a timeline or there could be a model where there's 10 decades and then Each decade has 10 points and so it's kind of like you could have um th thinner models that are nested structurally so that like the tree sweeps out all the possibilities still and so there's multiple different kinds of like hierarchical models that could be built um you could have you know 11s within a 20 within a 30 is like that doesn't make sense maybe it's not the most statistically optimal but that's that's not outside the space of statistical models that you can build because these are just statistical models however also people have gone I think a lot further in arguing for like time phenomenology

um an

example here's from 2018 an interview with Carl Friston from Alias um with Martin

forer so there's a question about that temporal

depth so we ask basically well you're highlighting temporal depth a lot like whether like thickness through time so is it enough for a computer program to do it and then if an ant colony is able to persist over longer time scales than other things like it then doesn't it have more thickness to

um it returns to this kind

of he brings up a lot of different components and I think it it like kind of remains to a very careful analysis like which which claims he he brings out

to respond to it um because it includes like questions about meta awareness the depth of the model in terms of the counterfactuals like for example like um use economic model the idea of of time discounting for example and financial modeling the idea that a \$100 today has more value than that \$100 10 years out because of properties like inflation for example um so yeah absolutely there's that kind of self-referencing quality like kind of Agents having self-awareness question but also this this temporal time domain has kind of external reality qualities if we consider time preferencing uh of value itself being a variable here or being an area of consideration yeah I think the deeper um the the the deeper philosophy is to be explored here's what it looks like in the code policy length equals four that's the time depth of that agent for time steps like that lets you estimate how many um how big the policy Vector is going to be if there's four times uh if there's four um affordance is or let's say five affordances like up down left right and weight at each time step then policy length of four is just every sequence of those five affordances so five to the 4th and then that's like tree search for policy just like early chess algorithms and then there's all these methods that that are that are different but the

starting point is just like the model
defines the temporal depth of the
plan now what does this say about the
the actuality of time that's probably
pretty

open thank you yeah thank thank you for
the question yeah and and then like that
is what motivates all these okay well
now how about inference on policy length
and all these other things like it's
just about starting with a model
iterating with it seeing what phenomena
of the real system still AR are
unaccounted that that gets
far

okay in last minutes any
questions okay let's see so that was the
first discussion yeah Eric I'm sorry uh
this this might be a little
straightforward to some but whenever we
were talking about $\dot{x}$ whenever we
were talking about the Dynamics of
internal external as well as boundary
state

the dot did that mean with respect to
the derivative of the actual boundary uh
like variables or was that with respect
to time now now I'm kind of questioning
myself great

question if it's no explain it
explain oh I was just going to say if
that would take us over time we can uh
discuss it afterward in the uh
questions yes I I think at a first
reading the dot on these are the the
derivative of the variable so that the
value at when it's zero that is a
stationerity of that

variable so but the Dynamics of the
variable are defined in terms of a
function of those other states plus a
noise parameter which is what makes it
amenable to like all these kind of
stochastic calculus physics approaches
because this like separation into a
signal in a noise term on a flow is the
basis of a lot of other
formalisms and it's always hard to also
I think jump between this is like the
physics
formalism like physics of particular
systems flows L Gran all of that but
then when we're actually doing the
modeling in discret
setting it's like where's the marov
blanket
here because here it's just a partially
observable marov decision process but
people have had that for decades so
clearly they didn't need the particular
physics to do a PDP
so that's kind of an interesting thing
to explore
too okay next week at the later
time second part of
three otherwise in the other cohort
we're in chapter eight which is
continuous time models but seven is on
discreet
time so thank you all see you next
time bye